

FERDOWS, Iran, Islamic Republic of

WMO: 407920

Lat: 34.0306N

Lon: 58.1834E

Elev: 4242

StdP: 12.58

Time Zone: 3.50 (IRN)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 19.0	(c) 24.9	(d) 0.6	(e) 6.6	(f) 38.0	(g) 4.8	(h) 8.2	(i) 40.8	(j) 20.2	(k) 51.0	(l) 17.7	(m) 49.0	(n) 3.8	(o) 270	(p) 0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	101.5	62.8	99.3	61.8	97.2	61.1	64.9	96.0	63.6	94.6	62.5	93.2	9.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.8	69.6	73.6	50.7	64.2	73.5	48.9	60.0	73.6	32.3	96.6	31.1	94.6	30.3	93.4	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.1	(b) 16.0	(c) 14.2	(e) 18.5	(f) 105.8	(g) 3.2	(h) 2.2	(i) 16.2	(j) 107.4	(k) 14.4	(l) 108.7	(m) 12.6	(n) 110.0	(o) 10.3	(p) 111.6		
(a) 18.1	(b) 16.0	(c) 14.2	(e) 13.7	(f) 66.7	(g) 6.0	(h) 2.8	(i) 9.4	(j) 68.7	(k) 5.8	(l) 70.3	(m) 2.5	(n) 71.9	(o) -1.9	(p) 73.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.7	40.3	45.0	55.7	65.9	76.2	84.6	86.6	82.4	76.1	66.1	52.4	44.2		
	DBStd	17.33	8.60	8.46	8.05	7.21	5.56	4.61	3.60	4.02	4.63	6.71	7.56	6.91		
	HDD50	787	311	175	37	2	0	0	0	0	0	0	59	203		
	HDD65	2799	764	560	301	73	2	0	0	0	1	70	382	646		
	CDD50	6167	11	35	215	478	813	1037	1136	1005	784	499	131	23		
	CDD65	2703	0	0	14	99	350	587	671	540	334	104	4	0		
	CDH74	34754	0	5	123	906	3867	8262	9788	6960	3772	1037	34	0		
	CDH80	18448	0	0	28	245	1714	4815	5863	3748	1730	303	2	0		
(14) Wind	WSAvg	6.5	5.2	5.9	6.8	6.5	7.1	7.3	8.1	7.7	6.6	6.2	5.3	5.1		
(15) Precipitation	PrecAvg	6.6	1.3	1.3	1.4	0.8	0.3	0.0	0.0	0.0	0.0	0.1	0.4	0.9		
	PrecMax	10.7	3.0	3.0	4.7	3.3	1.7	0.2	0.0	0.2	0.2	1.3	1.4	2.8		
	PrecMin	3.4	0.4	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.9	0.7	0.8	1.1	0.7	0.3	0.0	0.0	0.0	0.0	0.2	0.4	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.8	72.4	83.9	89.9	96.8	104.4	104.0	102.9	97.1	90.5	78.2	67.2		
		MCWB	45.0	49.2	53.4	57.5	59.1	64.6	63.7	63.1	59.2	57.7	53.0	47.3		
	2%	DB	60.9	65.7	76.9	86.0	94.0	101.4	101.7	98.3	93.7	86.4	73.0	62.9		
		MCWB	43.9	46.5	51.1	56.7	59.3	62.6	62.9	61.6	59.0	56.3	51.1	44.6		
	5%	DB	57.3	61.8	72.7	82.6	91.5	99.1	99.8	95.6	91.3	83.4	69.1	59.9		
		MCWB	42.5	45.0	50.3	55.7	59.0	61.7	62.2	60.5	58.2	55.5	49.4	43.3		
	10%	DB	53.6	58.3	69.4	79.4	88.9	96.8	97.6	93.4	89.0	80.3	65.5	56.3		
		MCWB	40.6	43.7	49.3	54.7	58.1	60.8	61.6	59.3	57.5	54.4	48.2	42.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.7	51.7	56.7	60.5	62.5	69.1	66.3	66.1	63.6	59.3	55.2	48.8		
		MCDB	57.8	65.4	68.4	80.7	91.5	99.2	98.7	93.6	90.8	86.6	71.6	61.2		
	2%	WB	45.8	49.2	54.3	58.5	61.0	64.8	64.8	63.8	61.3	57.6	52.8	46.8		
		MCDB	58.2	60.9	70.1	80.1	89.0	95.9	97.2	93.3	88.9	84.1	68.2	58.9		
	5%	WB	43.6	47.3	52.4	57.1	59.9	62.9	63.8	62.4	59.6	56.1	51.1	45.0		
		MCDB	55.2	58.2	69.1	78.8	86.9	94.5	95.7	91.9	88.2	81.1	65.6	56.8		
	10%	WB	41.4	45.4	50.6	55.7	58.9	61.8	62.8	60.9	58.1	54.8	49.4	43.2		
		MCDB	51.5	56.1	66.9	75.8	85.2	93.0	94.1	90.5	87.0	78.2	62.9	54.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	18.0	19.6	20.2	20.5	21.6	20.7	21.4	24.6	22.8	20.0	18.9		
		MCDBR	21.7	22.2	22.7	22.7	22.5	23.7	23.2	22.9	26.2	25.0	23.0	22.0		
		MCWBR	11.7	11.3	10.0	9.3	8.0	8.1	7.8	7.9	10.7	11.1	11.8	11.8		
	5% WB	MCDBR	18.9	18.4	19.4	20.9	21.4	22.7	21.6	21.5	25.7	24.2	20.4	18.6		
		MCWBR	11.1	10.1	9.3	8.9	8.0	8.1	7.8	7.7	11.1	11.0	10.9	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.317	0.355	0.425	0.477	0.488	0.463	0.456	0.417	0.373	0.381	0.348	0.310		
	taud		2.355	2.223	1.995	1.897	1.884	1.954	1.986	2.096	2.202	2.163	2.268	2.398		
	Ebn at Noon		282	280	267	259	257	263	264	272	281	266	266	278		
	Edh at Noon		31	39	53	62	63	59	57	50	43	41	33	28		
(44) All-Sky Solar Radiation	RadAvg		978	1257	1640	1994	2317	2613	2551	2383	2023	1554	1129	921		
	RadStd		90	95	76	102	104	67	38	67	42	49	67	71		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.83	N/A	-3.62	+1.42	N/A	-1.24	N/A	N/A	N/A	N/A

Nomenclature: See separate page